

7.7 Polymerisation (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.7 Polymerisation (A Level Only)
Difficulty	Easy

Time allowed: 30

Score: /20

Percentage: /100

Question 1a

This question is about condensation polymers.

State the names of **two** classes of organic molecules that can take part in condensation polymerisation.

[2 marks]

Question 1b

Draw the structure of the dicarboxylic acid whose formula is $C_4H_6O_4$. The functional groups should be displayed clearly.

[1 mark]

Question 1c

State the IUPAC name of the dicarboxylic acid in part (b)

[1 mark]

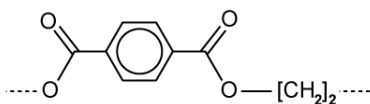
Question 1d

Draw one repeat unit of the polymer formed when $C_4H_6O_4$ polymerises with diethylamine, $H_2NCH_2CH_2NH_2$.

[2 marks]

Question 2a

The structure of a synthetic polyester is shown below.



Deduce the structures of two monomers used to make this polyester.

[2 marks]

Question 2b

One of the monomers is called benzene-1,4-dicarboxylic acid. State the name of the other one.

[1 mark]

Question 2c

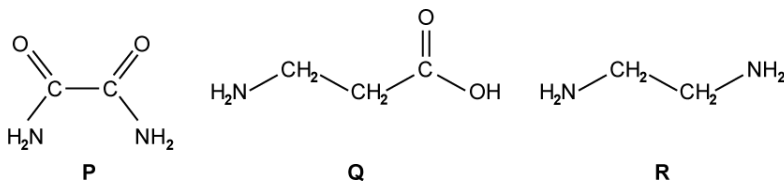
Name the other product of the reaction between the two monomers in part (b).

[1 mark]

Question 2d

Benzene-1,4-dicarboxylic acid will also react to form a polyamide.

Which of the three molecules could react with benzene-1,4-dicarboxylic acid to form a polyamide?



[1 mark]

Question 3a

This question is about hydrolysis of polymers.

What is meant by a hydrolysis reaction?

[1 mark]

Question 3b

Describe the reaction conditions and state what occurs to the polyester during acid hydrolysis.

[2 marks]

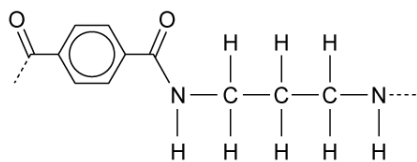
Question 3c

Outline the difference in the product(s) between acid and base hydrolysis of a polyester.

[2 marks]

Question 3d

Draw structures for the products of acid and base hydrolysis of the following polymer:



[4 marks]



Model Answers: Easy

1a

a) The names of **two** classes of organic molecules that can take part in condensation polymerisation are:

Any **two** from:

- dicarboxylic acids; [1 mark]
- hydrocarboxylic acids; [1 mark]
- diols; [1 mark]
- diamines; [1 mark]
- amino acids; [1 mark]
- dioyl chlorides; [1 mark]

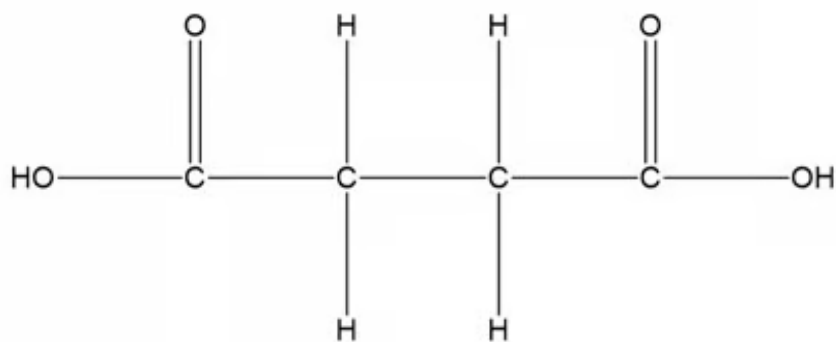
[Total: 2 marks]

- The molecules either to have two of the same functional group on either end of the monomer unit or two different functional groups that can react together
- The functional group combinations could be
 - carboxylic acid (or acyl chloride) + alcohol → ester + water
 - carboxylic acid (or acyl chloride) + amine → amide + water
 - amino acid → polyamide
- In the case of an amino acid, it can self-polymerise or react with different amino acids to make a condensation polymer since all amino acids contain the amine and carboxylic acid functional groups

1b

b) The structure of the dicarboxylic acid whose formula is $C_4H_6O_4$ is:

- Correct structure showing all the bonds; [1 mark]



[Total: 1 mark]

- The molecule is a dicarboxylic acid, meaning it has two -COOH groups
- Draw the carbon chain of four first and then complete the carboxylic acid groups, making sure one oxygen is double bonded
- Every carbon needs to have four bonds

1c

c) The IUPAC name of the dicarboxylic acid in part **(b)** is:

- (1,4-)butanedioic acid / butane-1,4-dioic acid; [1 mark]

[Total: 1 mark]

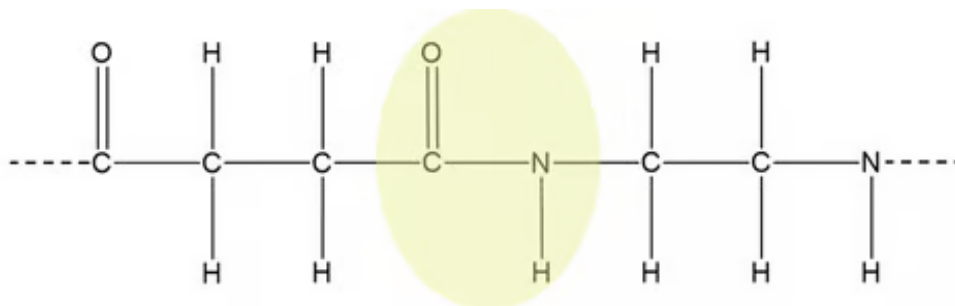
- The longest chain is four carbons counting the two carboxylic acid groups, so it must be derived from butane
- The pre-fix *di* goes in front of the *-oic acid* signifying the two carboxylic acid groups
- The IUPAC name does include the numbers 1,4 although the numbering is redundant as it is not possible to have the functional groups in any other position and still be a chain of four, so you would not be penalised in an exam if you left them out

1d

d) The repeat unit of the polymer formed when $\text{C}_4\text{H}_6\text{O}_4$ polymerises with diethylamine, $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$ is:

- Amide linkage (shown below); [1 mark]

- Rest of structure; [1 mark]



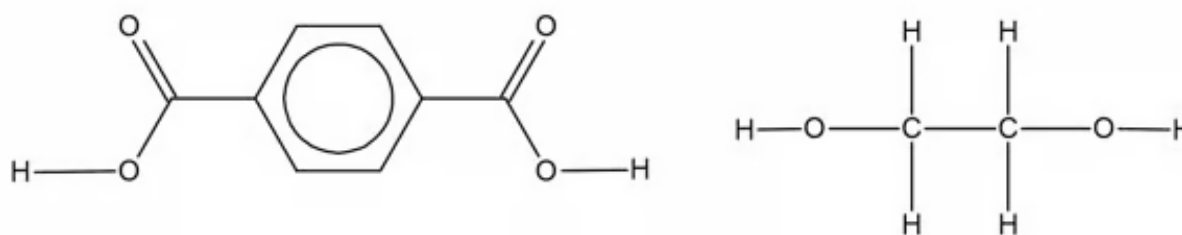
[Total: 2 marks]

- The amide link(age) is the name given to the join formed between a carboxylic acid and an amine group
- The repeat unit must end where the preceding segment begins
- For this polymer there must be four carbons and two carbons on either side of the nitrogen

2a

a) The two monomers used to make this polyester are:

- Correct structures; [2 marks]



[Total: 2 marks]

- To work how to draw the monomers correctly you have to pay attention to where to break the ester linkage
- The two oxygens will go together within the carboxylic acid group on either end of one monomer
- The other monomer can only have two carbons in since the repeat unit shows $(CH_2)_2$
- Complete the drawing of the second monomer by adding the alcohol groups

2b

b) The name of the other monomer is:

- Ethane-1,2-diol; [1 mark]

[Total: 1 mark]

- There are two alcohol groups so the compound is a diol
- You need the prefix *di* and the numbers to signify that the two alcohol groups are on separate carbons; if both groups were on the same carbon it would be called ethane-1,1-diol
- Diols and dicarboxylic acids are two important molecules used to make polyesters

2c

c) The other product of the reaction between the two monomers in part **(b)** is:

- Water; [1 mark]

[Total: 1 mark]

- Whenever a dicarboxylic acid and diol react together in a condensation reaction, water is the small molecule eliminated when the ester linkage is formed
- Condensation has nothing to do with water here; it is simply a reference to two molecules joining or condensing together

2d

d) The molecule which could react with benzene-1,4-dicarboxylic acid to form a polyamide is:

c) The other product of the reaction between the two monomers in part (b) is:

- Water; [1 mark]

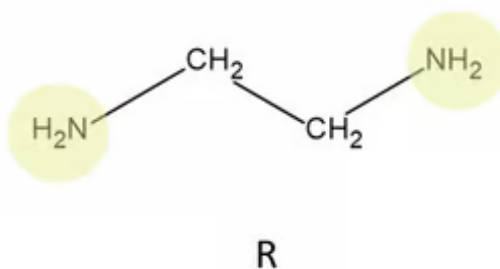
[Total: 1 mark]

- Whenever a dicarboxylic acid and diol react together in a condensation reaction, water is the small molecule eliminated when the ester linkage is formed
- Condensation has nothing to do with water here; it is simply a reference to two molecules joining or condensing together

2d

d) The molecule which could react with benzene-1,4-dicarboxylic acid to form a polyamide is:

- R; [1 mark]



[Total: 1 mark]

- A polyamide can be made from a dicarboxylic acid (or diol chloride) and a diamine
- The diamine is R as it has two amine groups (-NH_2) at either end of the chain which can react with carboxylic acid groups (or diol chlorides) to form a polyamide

3a

a) The meaning of a hydrolysis reaction is:

- The breakdown of a compound in aqueous solution or in the presence of water; [1 mark]

[Total: 1 mark]

- The word *hydro* should remind you of water and *lysis* means breaking down, as in electrolysis: breaking down with electricity

3b

b) What happens during the acid hydrolysis of a polyester:

- The polyester is heated (under reflux) in the presence of dilute acid / sulfuric acid / hydrochloric acid; [1 mark]
- The polyester breaks down with the acid acting as a catalyst; [1 mark]

[Total: 2 marks]

- The hydrolysis would be very slow without the presence of an acid or alkali and strong heating

3c

c) The difference in the product(s) between acid and base hydrolysis of a polyester is:

- Acid hydrolysis produces a dicarboxylic acid (and a diol); [1 mark]
- Base hydrolysis produces the salt of a dicarboxylic acid / dicarboxylate salt (and a diol); [1 mark]

[Total: 2 marks]

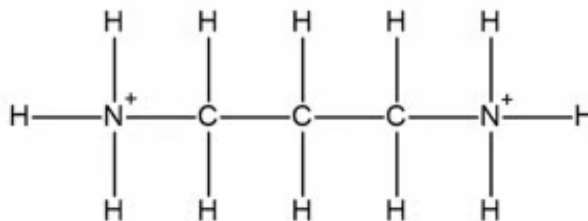
- The acid hydrolysis of an ester follows this pattern:
 - ester + water → carboxylic acid + alcohol
- Polyesters contain the ester linkage, -COO so the acid hydrolysis reaction produces the same products
- You just need to remember that the products have two of each functional group so it is a **dicarboxylic acid** and a **diol** there are produced

3d

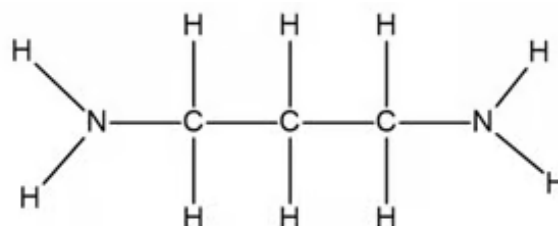
d) The products of acid and base hydrolysis of the polymer are:

- Correct structures; [4 marks]

Acid hydrolysis



Base hydrolysis



[Total: 4 marks]

- A polyamide is made from a dicarboxylic acid and diamine, however these are not the two products you get in acid and base hydrolysis
- Amines are basic groups, so in acid hydrolysis you will get the diamine salt or diamine cation along with the dicarboxylic acid following the pattern:
 - Amide + water → carboxylic acid + amine salt
- In base hydrolysis, the dicarboxylic acid groups will be neutralised and produce the dicarboxylate salt and diamine
 - Amide + water → carboxylate salt + amine